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(An Autonomous Institute Affiliated to Savitribai Phule University)

Explainable Automatic Modulation Recognition using CNN–GRU–GNN with Curriculum Learning and Retrieval-Augmented Generation

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Problem Statement

The Problem : What Is Currently Failing?

- Automatic Modulation Recognition (AMR) systems use deep learning
- Models behave as black-box classifiers
- Output only:
  Modulation label
  Confidence score
- No theoretical justification for predictions
- No adaptive recommendation based on channel conditions

The Consequence

- Engineers cannot check whether the AI decision follows communication theory.
- The model does not explain why a modulation was chosen.
- It does not show how noise affects BER or spectral efficiency.
- This reduces trust in AI systems used in telecom networks.
- Wrong modulation decisions can increase error rate and reduce system reliability.

Final Problem Statement

There is a need to develop an Automatic Modulation Recognition system that not only classifies modulation schemes accurately but also provides theory-grounded explanations and supports adaptive intelligence based on channel conditions.

Project Objectives

1. To develop an Automatic Modulation Recognition (AMR) system for accurate classification of modulation schemes from raw I/Q signal data

2. To capture spatial, temporal, and structural signal characteristics using a hybrid CNN–GRU–GNN architecture.

3. To improve robustness and performance under varying Signal-to-Noise Ratio (SNR) conditions using curriculum learning and fine-tuning.

4. To provide explainable predictions using communication-theoretic reasoning and Explainable AI techniques.

5. To enable adaptive modulation recommendation based on channel and SNR conditions.

Dataset & Signal Representation

~220K	**11**	**-20 to +18 dB**	**(2, 128)**
Samples	Modulation Classes	SNR Range	Input Shape

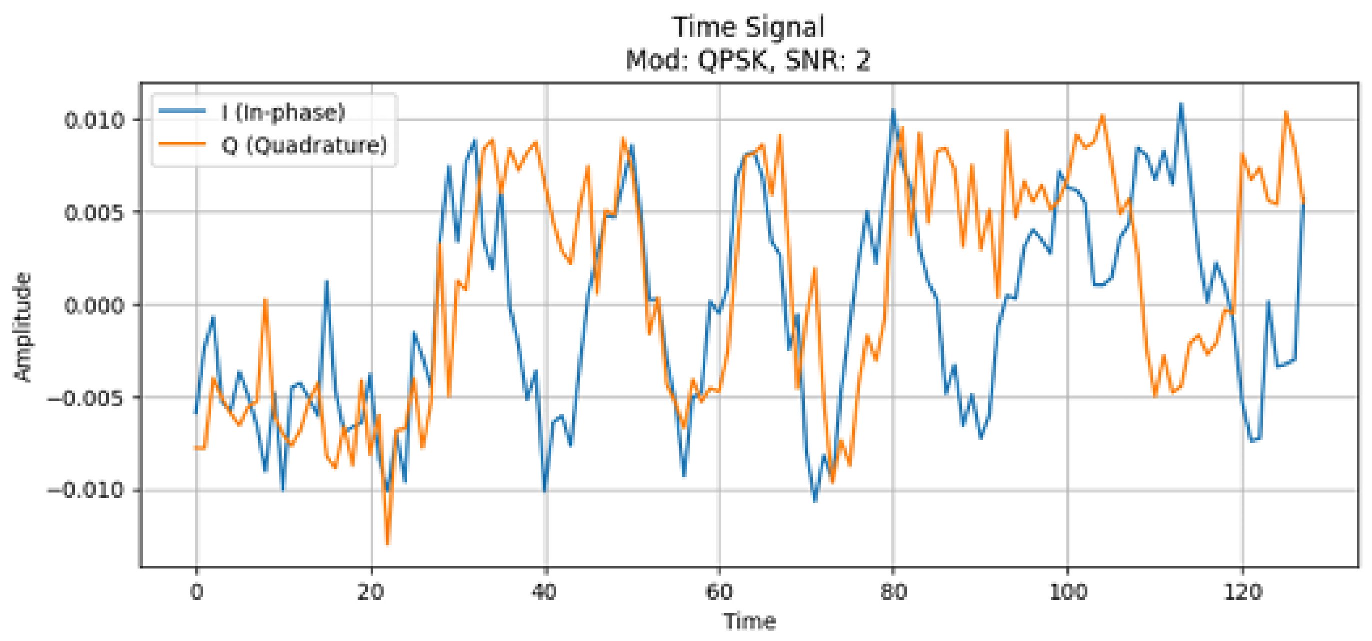


Fig. 2. Sample I/Q Signal Representation

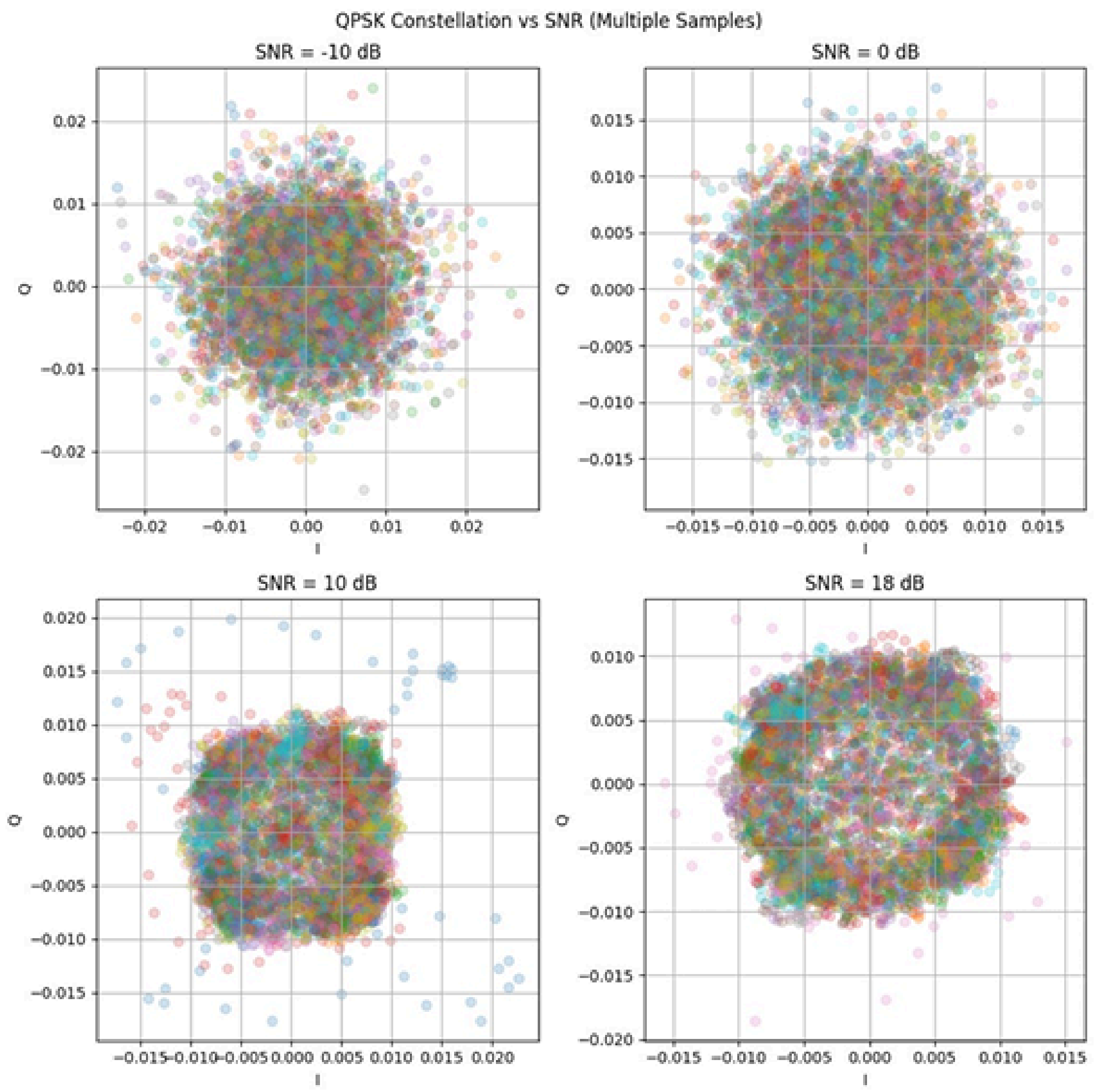
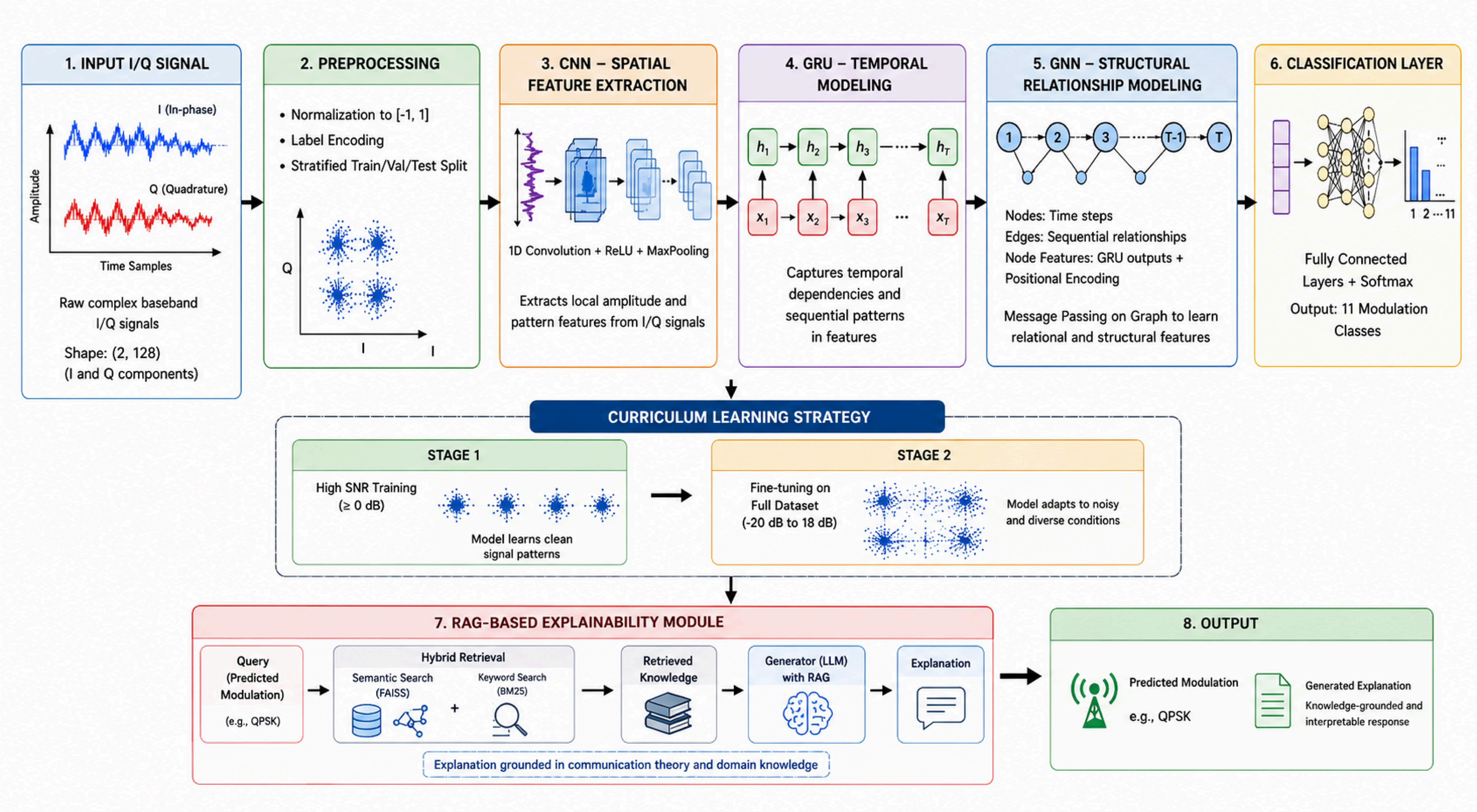


Fig. 1. Constellation Diagram under Different SNR Levels

⚙️ Preprocessing

- ✔ Signal normalization
- ✔ Label encoding
- ✔ Stratified train-test split
- ✔ Constellation analysis

System Architecture



Proposed CNN–GRU–GNN Architecture

I/Q Signal Input
(2, 128)

Preprocessing & Normalize
Label Encoding

CNN Spatial
Feature Extraction

GRU Temporal
Sequence Modeling

GNN Structural
Graph Relationships

Classifier Softmax
11 Classes

RAG Explain
Knowledge-Grounded

Component Responsibilities

Component	Role	Output	Key Benefit
CNN	Extracts local spatial patterns from I/Q signal	Feature maps	Amplitude & phase patterns
GRU	Models temporal dependencies in signal sequence	Hidden states	Efficient vs LSTM, no vanishing gradient
GNN	Learns relational structure across time steps	Node embeddings	Captures signal topology
RAG	Retrieves communication theory for explanation	Text explanation	Knowledge-grounded interpretability

Training Strategy & Explainability

Curriculum Learning Strategy

STAGE 1

High-SNR Training (≥ 0 dB)

93%
Accuracy

Model learns clean constellation patterns
Peak accuracy: ~93% on high-SNR samples

⬇️ **Fine-Tune on Full Dataset**

STAGE 2

Full Dataset Fine-Tuning (−20 to +18 dB)

Robust
Low-SNR

Model adapts to noisy low-SNR conditions
Improves generalization across all SNR levels

Explainable AI

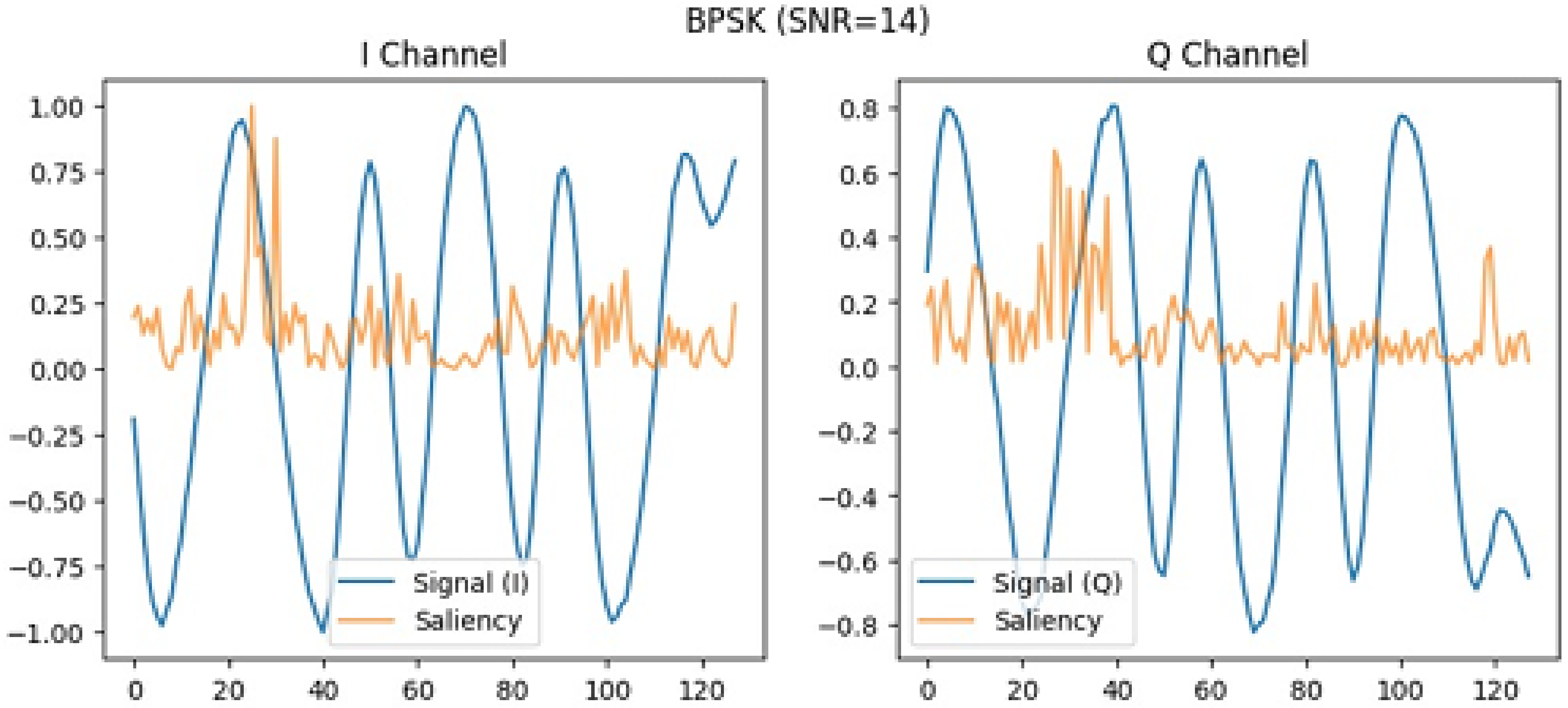
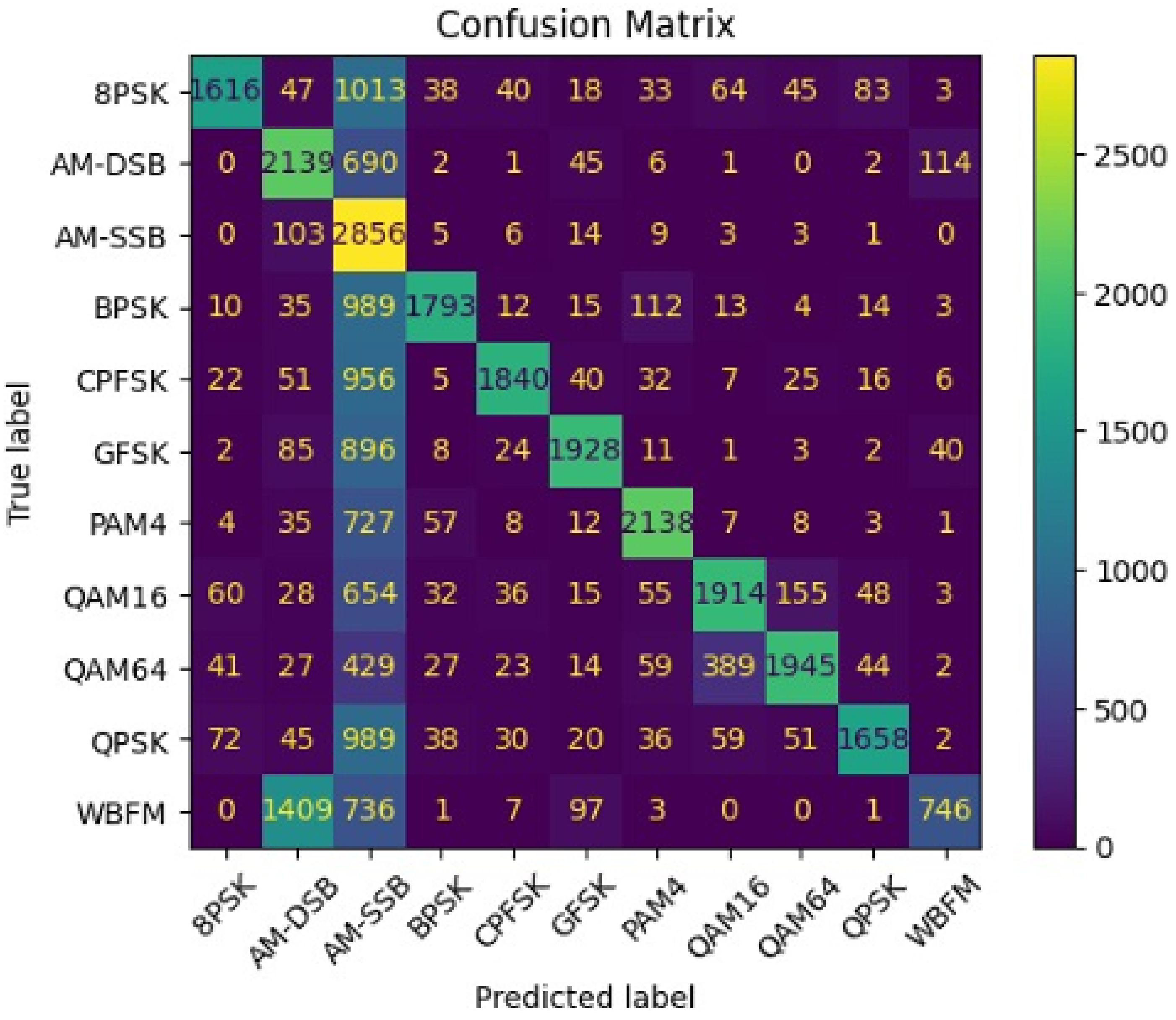


Fig. 4. Saliency Map Fine-Tuned Model

We first train using clean signals and then fine-tune using noisy data. Saliency maps help explain which signal regions affect the prediction.

Results & Performance Analysis

Model	Accuracy	Description
CNN	53%	Spatial feature extraction
CNN-LSTM	56%	Temporal modeling
CNN-GRU-GNN	93%	Proposed model



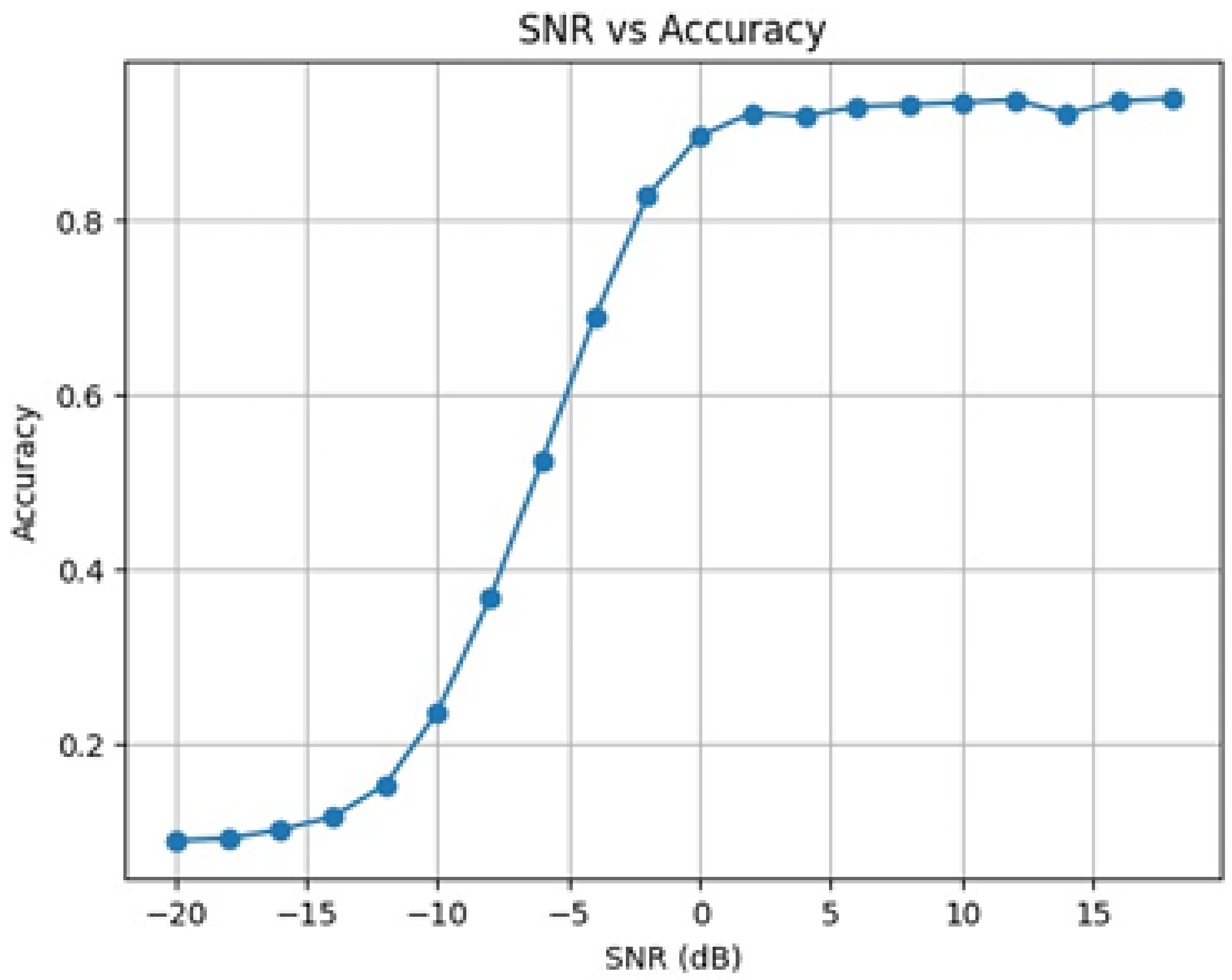


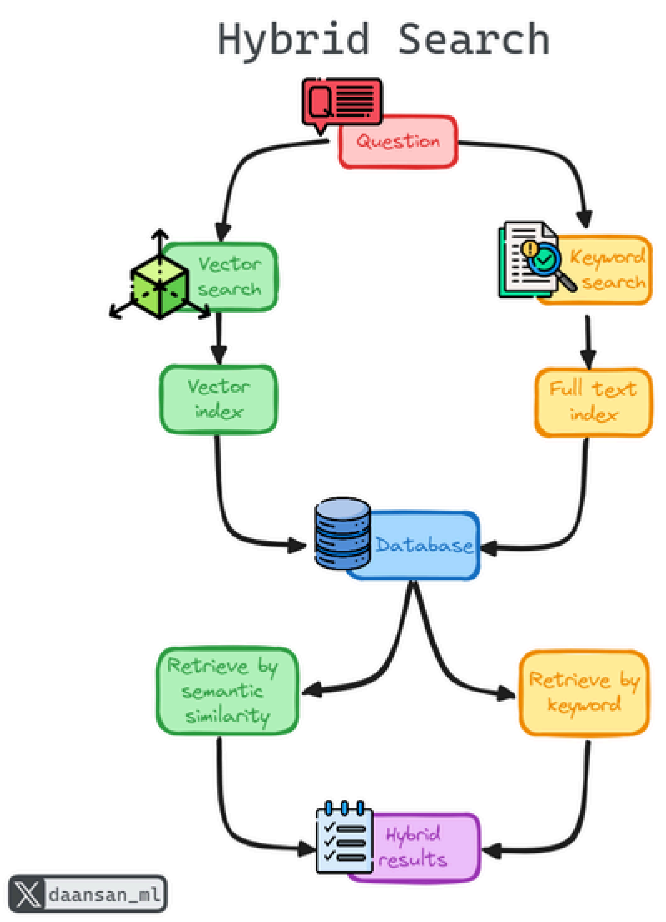
Fig. 6. SNR vs Accuracy for Fine-Tuned Model

Our proposed model significantly improves performance by combining spatial, temporal, and structural learning

Adaptive Intelligence & RAG Explainability

Adaptive Modulation Recommendation

SNR	Modulation	Description
SNR < 0 dB	**BPSK**	Maximum noise immunity Low data rate, highest reliability
SNR 0 – 8 dB	**QPSK**	Balanced performance Moderate noise handling
SNR 8 – 15 dB	**QAM16**	Higher throughput Good SNR conditions
SNR > 15 dB	**QAM64**	Maximum data rate Excellent channel quality



RAG Explainability Pipeline

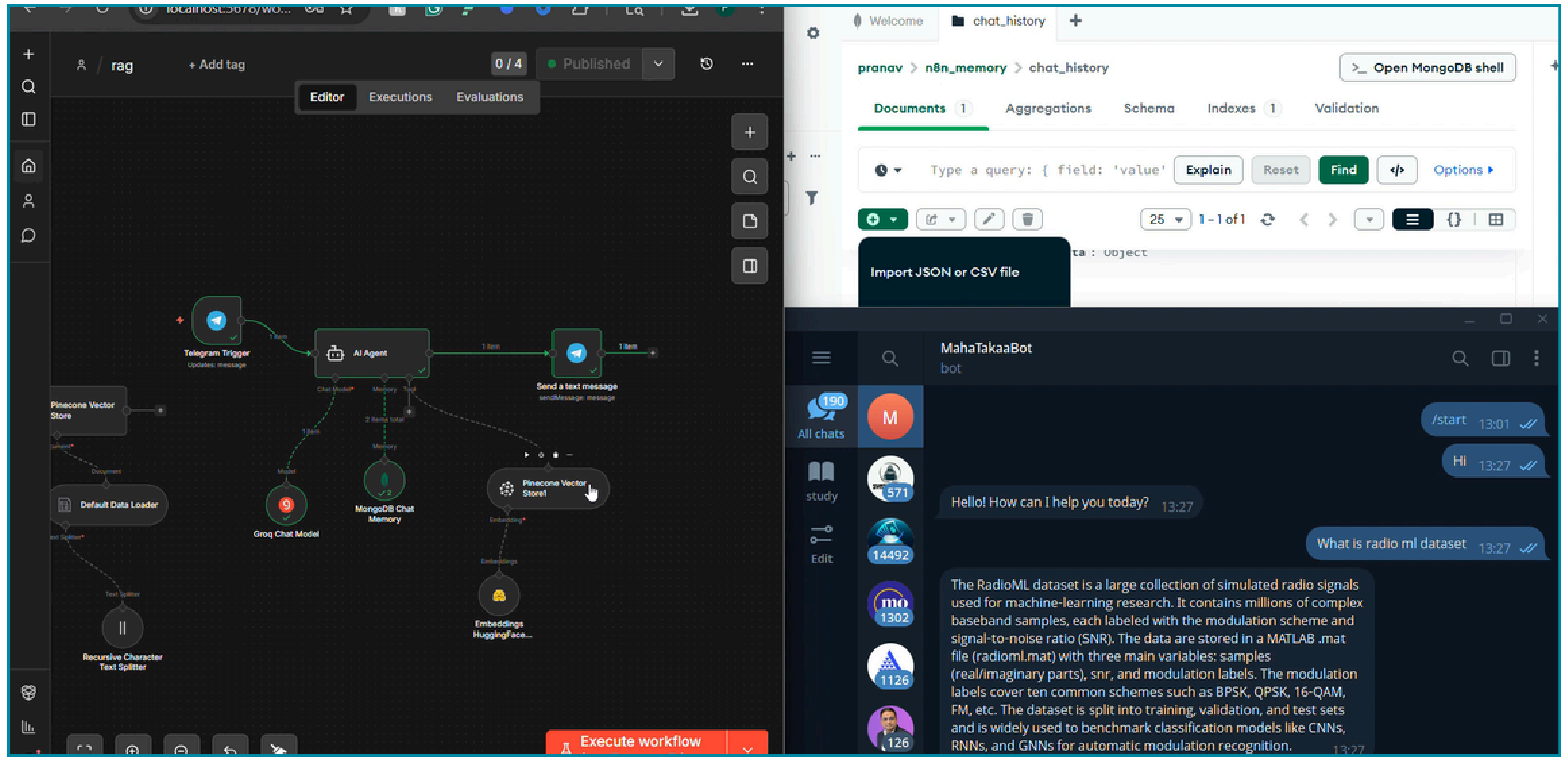
Query Formation
Predicted class + SNR level

Hybrid Retrieval
FAISS + BM25 Dual search

Retrieved Knowledge
Comm. theory documents

LLM Generator
Claude / GPT grounded gen.

Explanation Output
BER, noise & mod. rationale



Conclusion And Future Scope

Key Achievements

- 🏆 93% classification accuracy at high SNR conditions
- 🔬 Novel CNN–GRU–GNN hybrid for multi-level feature learning
- 🎓 Two-stage curriculum learning for low-SNR robustness
- 💡 RAG-based explainability grounded in communication theory
- 📡 Adaptive modulation recommendation for intelligent systems
- 📊 Outperforms CNN, CNN-LSTM, Transformer baselines

Future Scope

Graph Attention Networks (GAT)

Replace GNN with attention-weighted graph learning

Transformer Architectures

Investigate self-attention for signal classification

Real-time SDR Deployment

Implement on Software Defined Radio hardware

Edge & FPGA Implementation

Optimize for low-latency embedded systems

Reinforcement Learning

Dynamic adaptive modulation for real channels

THANK YOU